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NPTEL ONLINE
CERTIFICATION COURSE

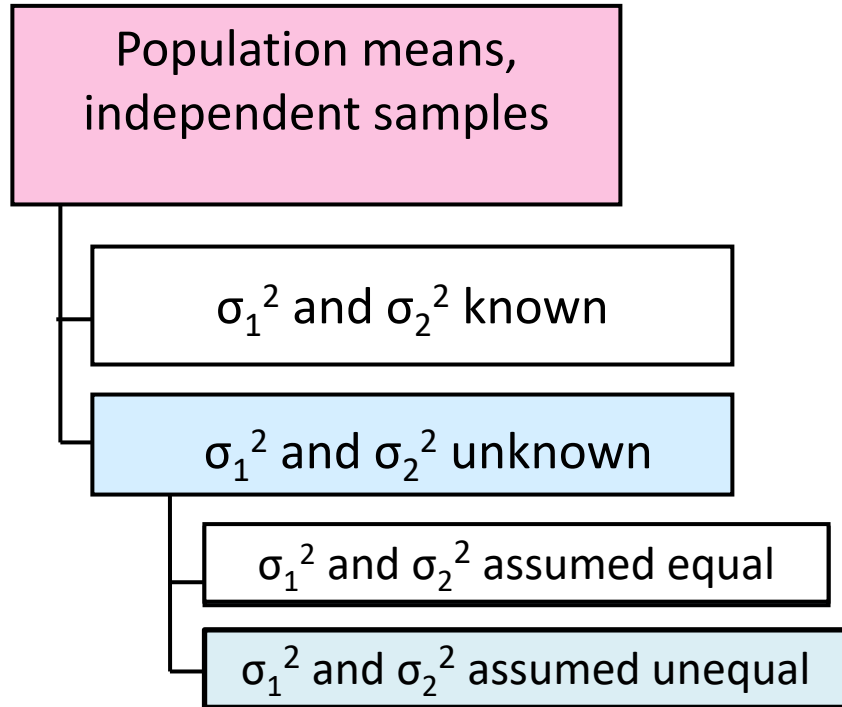
Hypothesis Testing: Two sample test

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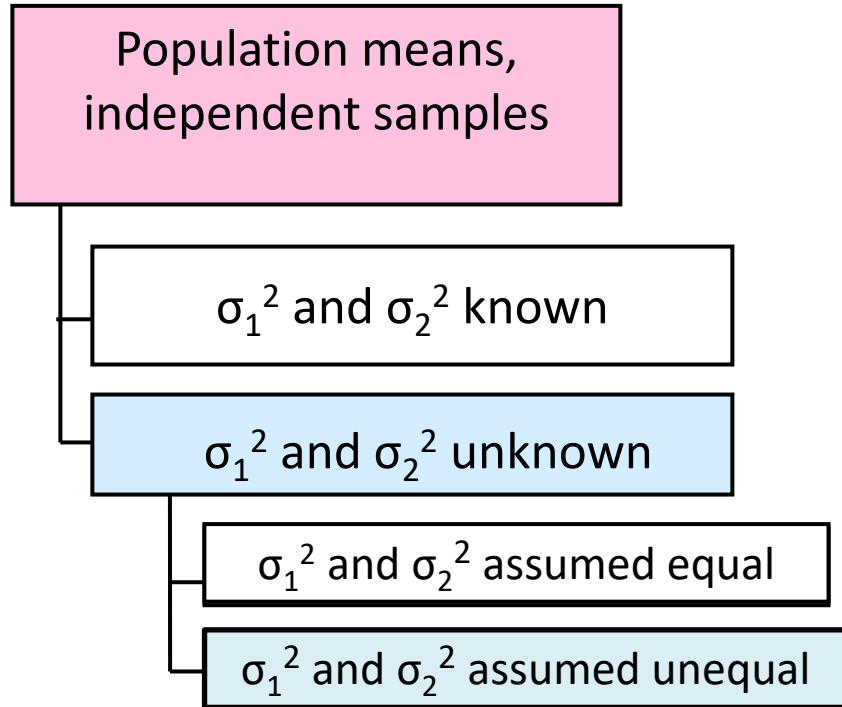
σ_1^2 and σ_2^2 Unknown, Assumed Unequal



Assumptions:

- Samples are randomly and independently drawn
- Populations are normally distributed
- Population variances are unknown and assumed unequal

σ_1^2 and σ_2^2 Unknown: Assumed Unequal



Forming interval estimates:

- The population variances are assumed unequal, so a pooled variance is not appropriate
- Use a **t value** with **v** degrees of freedom, where

$$v = \frac{\left[\left(\frac{s_1^2}{n_1} \right) + \left(\frac{s_2^2}{n_2} \right) \right]^2}{\left(\frac{s_1^2}{n_1} \right)^2 / (n_1 - 1) + \left(\frac{s_2^2}{n_2} \right)^2 / (n_2 - 1)}$$

Test Statistic: σ_1^2 and σ_2^2 Unknown, Unequal

σ_1^2 and σ_2^2 unknown

σ_1^2 and σ_2^2
assumed equal

σ_1^2 and σ_2^2
assumed unequal

The test statistic for

$\mu_1 - \mu_2$ is:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - D_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where t has v degrees of freedom:

$$v = \frac{\left[\left(\frac{s_1^2}{n_1} \right) + \left(\frac{s_2^2}{n_2} \right) \right]^2}{\left(\frac{s_1^2}{n_1} \right)^2 / (n_1 - 1) + \left(\frac{s_2^2}{n_2} \right)^2 / (n_2 - 1)}$$

Problem: Test Statistic: σ_1^2 and σ_2^2 Unknown, Unequal

- Arsenic concentration in public drinking water supplies is a potential health risk.
- An article in the Arizona Republic (Sunday, May 27, 2001) reported drinking water arsenic concentrations in parts per billion (ppb) for 10 metropolitan Phoenix communities and 10 communities in rural Arizona.
- The data as shown:

Metro Phoenix

Phoenix,	3
Chandler,	7
Gilbert,	25
Glendale,	10
Mesa,	15
Paradise Valley,	6
Peoria,	12
Scottsdale,	25
Tempe,	15
Sun City,	7

$$\bar{x}_1 = 12.5$$
$$s_1 = 7.63$$

Rural Arizona

Rimrock,	48
Goodyear,	44
New River,	40
Apache Junction,	38
Buckeye,	33
Nogales,	21
Black Canyon City,	20
Sedona,	12
Payson,	1
Casa Grande,	18

$$\bar{x}_1 = 27.5$$
$$s_2 = 15.3$$

Problem: Test Statistic: σ_1^2 and σ_2^2 Unknown, Unequal

- We wish to determine if there is any difference in mean arsenic concentrations between metropolitan Phoenix communities and communities in rural Arizona.

Problem: Test Statistic: σ_1^2 and σ_2^2 Unknown, Unequal

1. The parameters of interest are the mean arsenic concentrations for the two geographic regions, say, μ_1 and μ_2 , and we are interested in determining whether $\mu_1 - \mu_2 = 0$.
2. $H_0: \mu_1 - \mu_2 = 0$, or $H_0: \mu_1 = \mu_2$
3. $H_1: \mu_1 \neq \mu_2$
4. $\alpha = 0.05$ (say)
5. The test statistic is

$$t_0^* = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Problem: Test Statistic: σ_1^2 and σ_2^2 Unknown, Unequal

6. The degrees of freedom

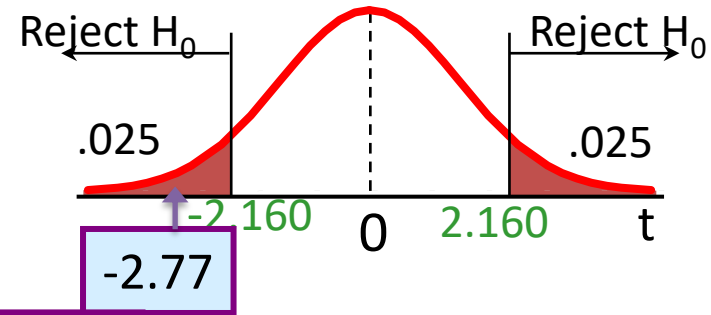
$$v = \frac{\left[\left(\frac{s_1^2}{n_1} \right) + \left(\frac{s_2^2}{n_2} \right) \right]^2}{\left(\frac{s_1^2}{n_1} \right)^2 / (n_1 - 1) + \left(\frac{s_2^2}{n_2} \right)^2 / (n_2 - 1)} = \frac{\left[\left(\frac{7.63^2}{10} \right) + \left(\frac{15.3^2}{10} \right) \right]^2}{\left(\frac{7.63^2}{10} \right)^2 / (10 - 1) + \left(\frac{15.3^2}{10} \right)^2 / (10 - 1)} = 13.2 = 13$$

Therefore, using $\alpha = 0.05$, we would reject $H_0: \mu_1 = \mu_2$ if $t_0^* > t_{0.025,13} = 2.160$ or if $t_0^* < -t_{0.025,13} = -2.160$

Problem: Test Statistic: σ_1^2 and σ_2^2 Unknown, Unequal

7. Computations:

$$t = \frac{(12.5 - 27.5) - 0}{\sqrt{\left(\frac{7.63^2}{10} + \frac{15.3^2}{10}\right)}} = -2.77$$



Decision:

Reject H_0 at $\alpha = 0.05$

Conclusion:

There is evidence of a difference in means.

Problem: Test Statistic: σ_1^2 and σ_2^2 Unknown, Unequal

8. Conclusions: Because $t_0^* = -2.77 < t_{0.025,13} = -2.160$,

- Reject the null hypothesis.
- There is evidence to conclude that mean arsenic concentration in the drinking water in rural Arizona is different from the mean arsenic concentration in metropolitan Phoenix drinking water.

Problem: Test Statistic: σ_1^2 and σ_2^2 Unknown, Unequal

```
In [17]: stats.t.ppf(0.025,13) #critical t value
```

```
Out[17]: -2.160368656461013
```

```
In [18]: metro = [3,7,25,10,15,6,12,25,15,7]
         rural = [48,44,40,38,33,21,20,12,1,18]
```

```
In [20]: stats.ttest_ind(metro,rural, equal_var = False)
```

```
Out[20]: Ttest_indResult(statistic=-2.7669395785560558, pvalue=0.015827284816100885)
```

Dependent Samples

Tests Means of 2 **Related** Populations

- **Paired or matched** samples
- Repeated measures (before/after)
- Use **difference** between paired values:

$$d_i = x_i - y_i$$

- Assumptions:
 - Both Populations Are Normally Distributed

Test Statistic: Dependent Samples

The test statistic for the mean difference is a **t value**, with **$n - 1$ degrees of freedom**:

$$t = \frac{\bar{d} - D_0}{\frac{s_d}{\sqrt{n}}}$$

$$\bar{d} = \frac{\sum d_i}{n} = \bar{x} - \bar{y}$$

D_0 = hypothesized mean difference

s_d = sample standard dev. of differences

n = the sample size (number of pairs)

Decision Rules: Dependent Samples

Lower-tail test:

$$H_0: \mu_1 - \mu_2 \geq 0$$

$$H_1: \mu_1 - \mu_2 < 0$$

Upper-tail test:

$$H_0: \mu_1 - \mu_2 \leq 0$$

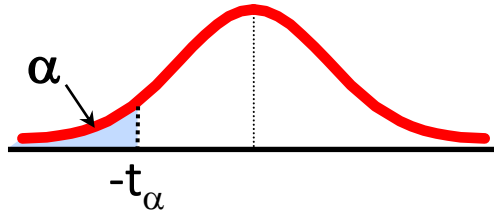
$$H_1: \mu_1 - \mu_2 > 0$$

Two-tail test:

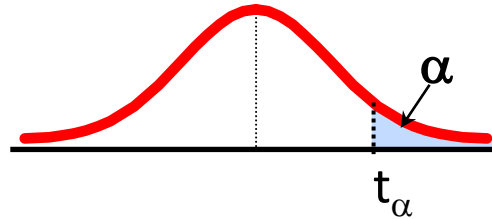
$$H_0: \mu_1 - \mu_2 = 0$$

$$H_1: \mu_1 - \mu_2 \neq 0$$

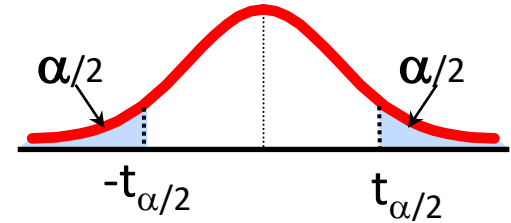
Decision Rules: Dependent Samples



Reject H_0 if $t < -t_{n-1, \alpha}$



Reject H_0 if $t > t_{n-1, \alpha}$



Reject H_0 if $t < -t_{n-1, \alpha/2}$
or $t > t_{n-1, \alpha/2}$

Where

$$t = \frac{\bar{d} - D_0}{\frac{s_d}{\sqrt{n}}}$$

has $n - 1$ d.f.

Dependent Samples: Example

- An article in the Journal of Strain Analysis (1983, Vol. 18, No. 2) compares several methods for predicting the shear strength for steel plate girders.
- Data for two of these methods, the Karlsruhe and Lehigh procedures, when applied to nine specific girders, are shown in Table .
- We wish to determine whether there is any difference (on the average) between the two methods.

Table : Strength Predictions for Nine Steel Plate Girders
(Predicted Load/Observed Load)

Girder	Karlsruhe Method	Lehigh Method	Difference dj
S11	1.186	1.061	0.119
S21	1.151	0.992	0.159
S31	1.322	1.063	0.259
S41	1.339	1.062	0.277
S51	1.200	1.065	0.138
S21	1.402	1.178	0.224
S22	1.365	1.037	0.328
S23	1.537	1.086	0.451
S24	1.559	1.052	0.507

Inferences About the Difference Between Two Population Means: Matched Samples

1. The parameter of interest is the difference in mean shear strength between the two methods, say, $\mu_D = \mu_1 - \mu_2 = 0$.
2. $H_0: \mu_D = 0$
3. $H_1: \mu_D \neq 0$
4. $\alpha = 0.05$
5. The test statistic is

$$t_0 = \frac{\bar{d}}{s_D/\sqrt{n}}$$

Inferences About the Difference Between Two Population Means: Matched Samples

6. Reject H_0 if $t_0 > t_{0.025,8} = 2.306$ or if $t_0 < -t_{0.025,8} = -2.306$.
7. Computations: The sample average and standard deviation of the differences d_j are $\bar{d} = 0.2736$ and $s_D = 0.1356$, so the test statistic is

$$t_0 = \frac{\bar{d}}{s_D/\sqrt{n}} = \frac{0.2736}{0.1356/\sqrt{9}} = 6.05$$

8. Conclusions: Since $t_0 = 6.05 > 2.306$,

we conclude that the strength prediction methods yield different results.

The P -value for $t_0 = 6.05$ is $P = 0.0002$,

```
In [37]: KARL= [1.186,1.151,1.322,1.339,1.200,1.402,1.365,1.537,1.559]  
        LEH= [1.061,0.992,1.063,1.062,1.065,1.178,1.037,1.086,1.052]
```

```
In [38]: stats.ttest_rel(KARL,LEH)
```

```
Out[38]: Ttest_relResult(statistic=6.0819394375848255, pvalue=0.00029529546278604066)
```

Inferences About the Difference Between Two Population Proportions

- Inferences About the Difference Between Two Population Proportion



Inferences About the Difference Between Two Population Proportions

- Interval Estimation of $p_1 - p_2$
- Hypothesis Tests About $p_1 - p_2$



Sampling Distribution of $\bar{p}_1 - \bar{p}_2$

- Expected Value

$$E(\bar{p}_1 - \bar{p}_2) = p_1 - p_2$$

- Standard Deviation (Standard Error)

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

where: n_1 = size of sample taken from population 1

n_2 = size of sample taken from population 2

Sampling Distribution of $\bar{p}_1 - \bar{p}_2$

- If the sample sizes are large, the sampling distribution of $\bar{p}_1 - \bar{p}_2$ can be approximated by a normal probability distribution.
- The sample sizes are sufficiently large if all of these conditions are met:

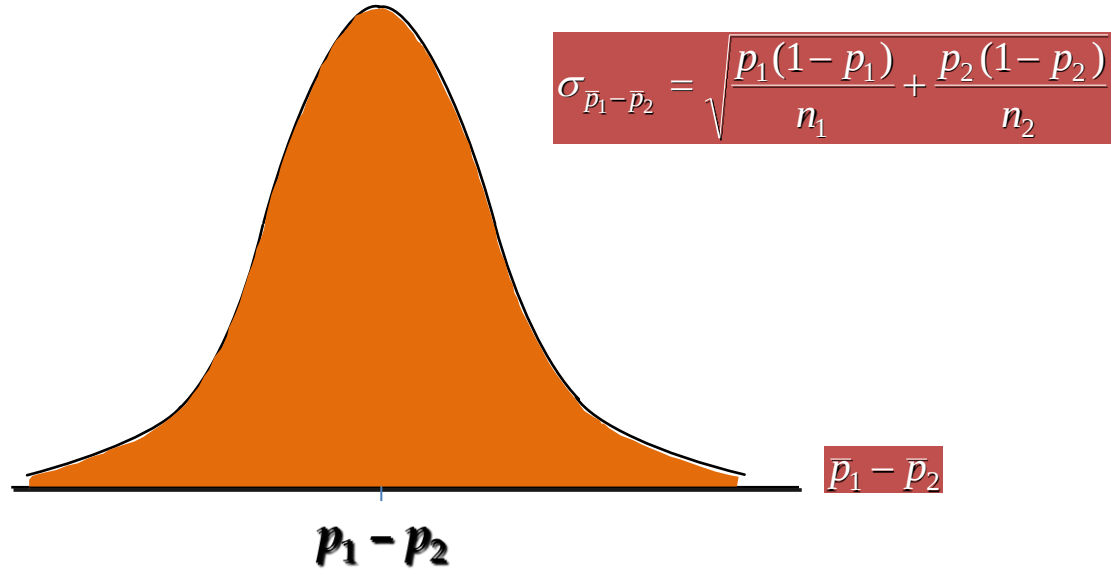
$$n_1 p_1 \geq 5$$

$$n_1 (1 - p_1) \geq 5$$

$$n_2 p_2 \geq 5$$

$$n_2 (1 - p_2) \geq 5$$

Sampling Distribution of $\bar{p}_1 - \bar{p}_2$



Interval Estimation of $p_1 - p_2$

- Interval Estimation

$$\bar{p}_1 - \bar{p}_2 \pm z_{\alpha/2} \sqrt{\frac{\bar{p}_1(1-\bar{p}_1)}{n_1} + \frac{\bar{p}_2(1-\bar{p}_2)}{n_2}}$$

Point Estimator of the Difference Between Two Population Proportions

- p_1 = proportion of the population of households “aware” of the product after the new campaign
- p_2 = proportion of the population of households “aware” of the product before the new campaign
- $\bar{p}_1$ = sample proportion of households “aware” of the product after the new campaign
- $\bar{p}_2$ = sample proportion of households “aware” of the product before the new campaign

$$\bar{p}_1 - \bar{p}_2 = \frac{120}{250} - \frac{60}{150} = .48 - .40 = .08$$

Hypothesis Tests about $p_1 - p_2$

- Hypothesis

We focus on tests involving no difference between the two population proportions (i.e. $p_1 = p_2$)

$$H_0: p_1 - p_2 \geq 0$$

$$H_a: p_1 - p_2 < 0$$

Left-tailed

$$H_0: p_1 - p_2 \leq 0$$

$$H_a: p_1 - p_2 > 0$$

Right-tailed

$$H_0: p_1 - p_2 = 0$$

$$H_a: p_1 - p_2 \neq 0$$

Two-tailed

Hypothesis Tests about $p_1 - p_2$

- Standard Error of $\bar{p}_1 - \bar{p}_2$ when $p_1 = p_2 = p$

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

- Pooled Estimator of p when $p_1 = p_2 = p$

$$\bar{p} = \frac{n_1 \bar{p}_1 + n_2 \bar{p}_2}{n_1 + n_2}$$

Hypothesis Tests about $p_1 - p_2$

- Test Statistic

$$z = \frac{(\bar{p}_1 - \bar{p}_2)}{\sqrt{\bar{p}(1 - \bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Problem: Hypothesis Tests about $p_1 - p_2$

- Extracts of St. John's Wort are widely used to treat depression.
- An article in the April 18, 2001 issue of the *Journal of the American Medical Association* ("Effectiveness of St. John's Wort on Major Depression: A Randomized Controlled Trial") compared the efficacy of a standard extract of St. John's Wort with a placebo in 200 outpatients diagnosed with major depression.
- Patients were randomly assigned to two groups; one group received the St. John's Wort, and the other received the placebo.
- After eight weeks, 19 of the placebo-treated patients showed improvement, whereas 27 of those treated with St. John's Wort improved.
- Is there any reason to believe that St. John's Wort is effective in treating major depression? Use $\alpha = 0.05$.



Problem: Hypothesis Tests about $p_1 - p_2$

1. The parameters of interest are p_1 and p_2 , the proportion of patients who improve following treatment with St. John's Wort (p_1) or the placebo (p_2).
2. $H_0: p_1 = p_2$
3. $H_1: p_1 \neq p_2$
4. $\alpha = 0.05$
5. The test statistic is

$$z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where $\hat{p}_1 = 27/100 = 0.27$, $\hat{p}_2 = 19/100 = 0.19$, $n_1 = n_2 = 100$, and

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{19 + 27}{100 + 100} = 0.23$$



Problem: Hypothesis Tests about $p_1 - p_2$

6. Reject $H_0: p_1 = p_2$ if $z_0 > z_{0.025} = 1.96$ or if $z_0 < -z_{0.025} = -1.96$.
7. Computations: The value of the test statistic is

$$z_0 = \frac{0.27 - 0.19}{\sqrt{0.23(0.77)\left(\frac{1}{100} + \frac{1}{100}\right)}} = 1.35$$

8. Conclusions: Since $z_0 = 1.35$ does not exceed $z_{0.025}$, we cannot reject the null hypothesis. The P -value is $P \cong 0.177$. There is insufficient evidence to support the claim that St. John's Wort is effective in treating major depression.

```
In [29]: import math
def two_samp_proportion(p1,p2,n1,n2):
    p_pool = ((p1*n2)+(p2*n1))/(n1+n2)
    x = (p_pool*(1- p_pool)*((1/n1)+(1/n2)))
    s = math.sqrt(x)
    z = (p1- p2)/s
    if (z < 0):
        p_val = stats.norm.cdf(z)
    else:
        p_val = 1 - stats.norm.cdf(z)
    return z, p_val*2
```

```
In [30]: two_samp_proportion(0.27,0.19,100,100)
```

```
Out[30]: (1.3442056254198995, 0.17888190308175567)
```

```
In [27]: stats.norm.cdf(1.3442056254198995)
```

```
Out[27]: 0.9105590484591222
```

Thank You





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Hypothesis Testing: Two sample test

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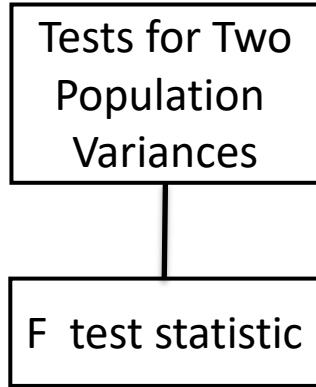


Agenda

- Comparing two population variances
- Choosing z or t test
- Sample size

Hypothesis Tests for Two Variances

Goal: Test hypotheses about two population variances



$$H_0: \sigma_1^2 \geq \sigma_2^2$$

$$H_1: \sigma_1^2 < \sigma_2^2$$

Lower-tail
test

$$H_0: \sigma_1^2 \leq \sigma_2^2$$

$$H_1: \sigma_1^2 > \sigma_2^2$$

Upper-tail
test

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

Two-tail test

The two populations are assumed to be independent and normally distributed

Hypothesis Tests for Two Variances

Tests for Two
Population
Variances

F test statistic

The random variable

$$F = \frac{s_1^2 / \sigma_1^2}{s_2^2 / \sigma_2^2}$$

Has an F distribution with $(n_1 - 1)$ numerator degrees of freedom and $(n_2 - 1)$ denominator degrees of freedom

Denote an F value with v_1 numerator and v_2 denominator degrees of freedom by

Test Statistic

Tests for Two
Population
Variances

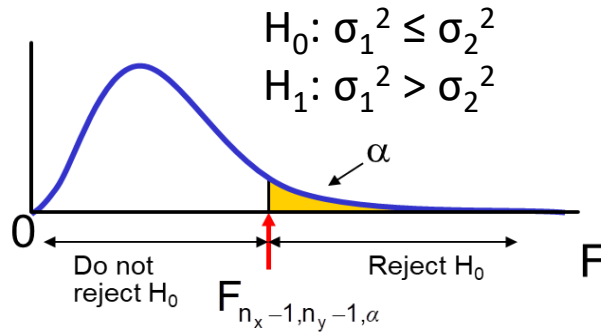
F test statistic

The critical value for a hypothesis test about two population variances is

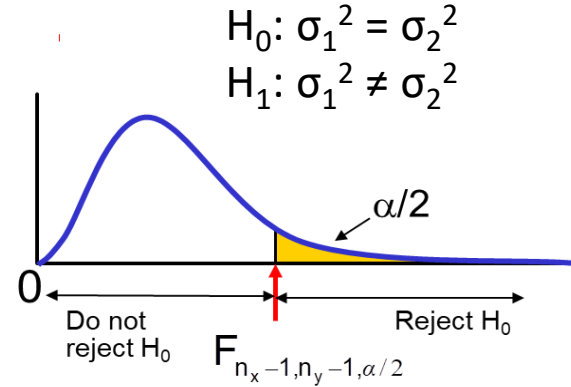
$$F = \frac{s_1^2}{s_2^2}$$

where F has $(n_x - 1)$ numerator degrees of freedom and $(n_y - 1)$ denominator degrees of freedom

Decision Rules: Two Variances



Reject H_0 if $F > F_{n_x-1, n_y-1, \alpha}$



■ rejection region for a two-tail test is:

Reject H_0 if $F > F_{n_x-1, n_y-1, \alpha/2}$

where s_x^2 is the larger of the two sample variances

Problem

- A company manufactures impellers for use in jet-turbine engines.
- One of the operations involves grinding a particular surface finish on a titanium alloy component.
- Two different grinding processes can be used, and both processes can produce parts at identical mean surface roughness.
- The manufacturing engineer would like to select the process having the least variability in surface roughness.
- A random sample of $n_1 = 11$ parts from the first process results in a sample standard deviation $s_1 = 5.1$ micro inches, and a random sample of $n_2 = 16$ parts from the second process results in a sample standard deviation of $s_2 = 4.7$ micro inches.
- We will find a 90% confidence interval on the ratio of the two standard deviations.

Problem

- Form the hypothesis test:
 $H_0: \sigma_1^2 = \sigma_2^2$ (there is no difference between variances)
 $H_1: \sigma_1^2 \neq \sigma_2^2$ (there is a difference between variances)
- Find the F critical values for $\alpha = .10/2$:

Degrees of Freedom:

- Numerator
 - $n_1 - 1 = 11 - 1 = 10$ d.f.
- Denominator:
 - $n_2 - 1 = 16 - 1 = 15$ d.f.

Problem

- Assuming that the two processes are independent and that surface roughness is normally distributed

$$\frac{s_1^2}{s_2^2} f_{0.95,15,10} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} f_{0.05,15,10}$$
$$\frac{(5.1)^2}{(4.7)^2} 0.39 \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{(5.1)^2}{(4.7)^2} 2.85$$

or upon completing the implied calculations and taking square roots,

$$0.678 \leq \frac{\sigma_1}{\sigma_2} \leq 1.887$$

Table of F-statistics $P=0.05$

[t-statistics](#)

F-statistics with other P-values: [P=0.01](#) | [P=0.001](#)

[Chi-square statistics](#)

df2\df1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.76	8.74	8.73	8.71	8.70
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.94	5.91	5.89	5.87	5.86
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.70	4.68	4.66	4.64	4.62
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.03	4.00	3.98	3.96	3.94
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.60	3.57	3.55	3.53	3.51
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.31	3.28	3.26	3.24	3.22
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.10	3.07	3.05	3.03	3.01
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.94	2.91	2.89	2.86	2.85
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.82	2.79	2.76	2.74	2.72
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.72	2.69	2.66	2.64	2.62
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.63	2.60	2.58	2.55	2.53
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.57	2.53	2.51	2.48	2.46
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.51	2.48	2.45	2.42	2.40

Problem

- $f_{0.95,15,10} = 1 / f_{0.05,10,15} = 1/2.54 = 0.39$
- Since this confidence interval includes unity, we cannot claim that the standard deviations of surface roughness for the two processes are different at the 90% level of confidence.

```
In [1]: import pandas as pd
import numpy as np
import math
from scipy import stats
import scipy
```

```
In [45]: scipy.stats.f.ppf(q=1-0.05, dfn= 15, dfd=10)|
```

```
Out[45]: 2.8450165269958436
```

```
In [44]: scipy.stats.f.ppf(q=0.05, dfn=15, dfd=10)
```

```
Out[44]: 0.3931252536255495
```


F Test example:

```
In [9]: X = [3,7,25,10,15,6,12,25,15,7]
        Y = [48,44,40,38,33,21,20,12,1,18]
        import numpy as np
```

```
In [11]: F = np.var(X) / np.var(Y)
        dfn = len(X) - 1
        dfd = len(Y) - 1
```

```
In [12]: p_value = scipy.stats.f.cdf(F, dfn, dfd)
```

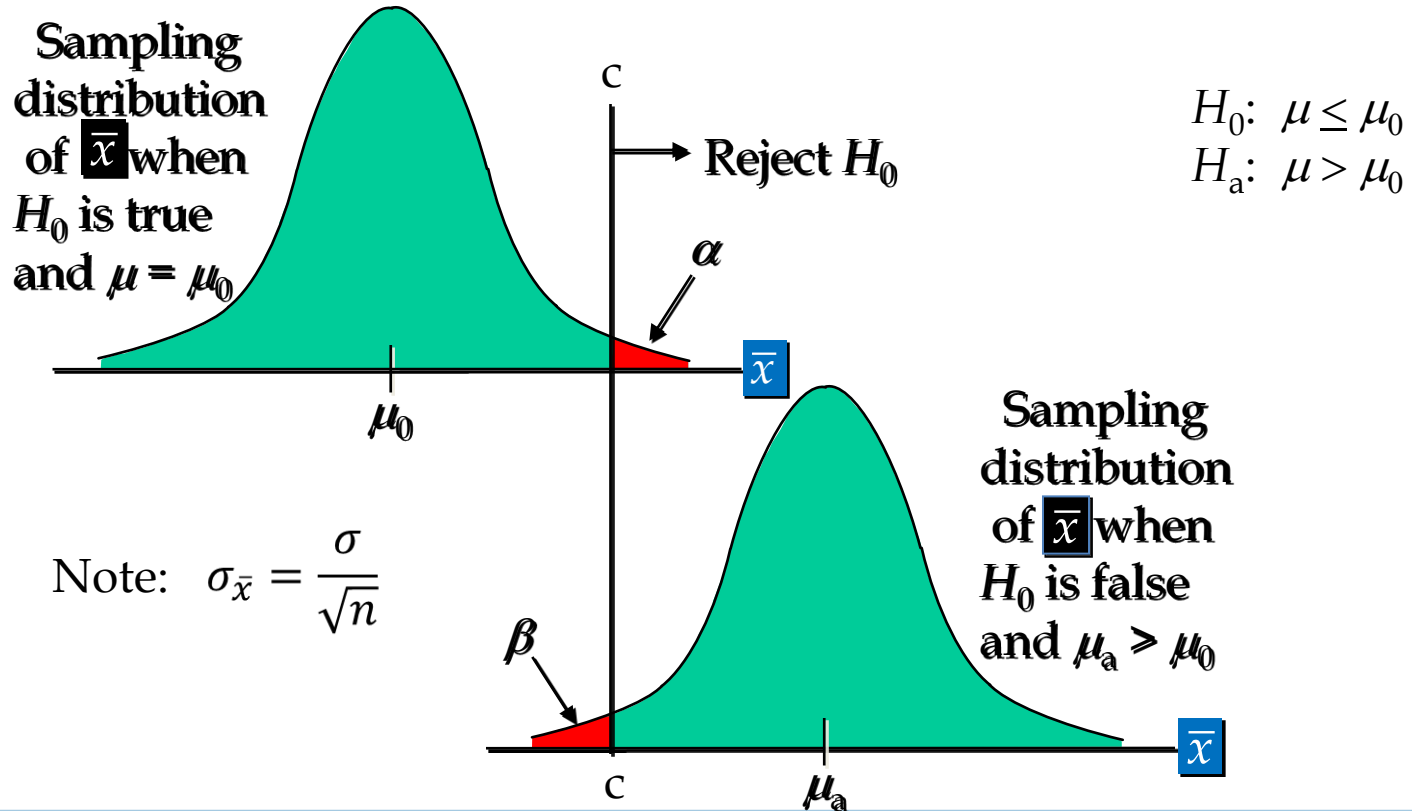
```
In [13]: p_value
```

```
Out[13]: 0.024680183438910465
```

Z Vs t

	σ –known	σ –unknown
$n \leq 30$	Z-test	t-test
$n > 30$	Z-test	Z-test Use Sample standard deviation

Determining the Sample Size for a Hypothesis Test About a Population Mean



Determining the Sample Size for a Hypothesis Test About a Population Mean

$$n = \frac{(z_{\alpha} + z_{\beta})^2 \sigma^2}{(\mu_0 - \mu_a)^2}$$

where

z_{α} = z value providing an area of α in the tail

z_{β} = z value providing an area of β in the tail

σ = population standard deviation

μ_0 = value of the population mean in H_0

μ_a = value of the population mean used for the
Type II error

Note: In a two-tailed hypothesis test, use $z_{\alpha/2}$ not z_{α}



Determining the Sample Size for a Hypothesis Test About a Population Mean

- Let's assume that the manufacturing company makes the following statements about the allowable probabilities for the Type I and Type II errors:
- If the mean diameter is $\mu = 12$ mm, I am willing to risk an $\alpha = .05$ probability of rejecting H_0 .
- If the mean diameter is 0.75 mm over the specification ($\mu = 12.75$), I am willing to risk a $\beta = .10$ probability of not rejecting H_0 .



Determining the Sample Size for a Hypothesis Test About a Population Mean

$$\alpha = .05, \beta = .10$$

$$z_{\alpha} = 1.645, z_{\beta} = 1.28$$

$$\mu_0 = 12, \mu_a = 12.75$$

$$\sigma = 3.2$$

$$\begin{aligned} n &= \frac{(z_{\alpha} + z_{\beta})^2 \sigma^2}{(\mu_0 - \mu_a)^2} = \frac{(1.645 + 1.28)^2 (3.2)^2}{(12 - 12.75)^2} \\ &= 155.75 \approx 156 \end{aligned}$$

```
In [2]: import pandas as pd
import numpy as np
from scipy import stats
```

```
In [5]: import math
```

```
In [22]: def samplesize(alfa,beta,mu1,mu2,sigma):
z1 = -1*stats.norm.ppf(alfa)
z2 = -1*stats.norm.ppf(beta)
n= (((z1+z2)**2)*(sigma**2))/((mu1-mu2)**2)
print (n)
```

```
In [23]: samplesize(0.05,0.1,12,12.75,3.2)
```

155.900083325938

Thank You





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ANOVA

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Agenda

- Sample Size Calculation
- One Way ANOVA – Introduction

Determining Sample Size when Estimating μ

- Z formula**
$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

- Error of Estimation (tolerable error)**
$$E = \bar{X} - \mu$$

- Estimated Sample Size**
$$n = \frac{Z_{\frac{\alpha}{2}}^2 \sigma^2}{E^2} = \left(\frac{Z_{\frac{\alpha}{2}} \sigma}{E} \right)^2$$

- Estimated σ**
$$\sigma \approx \frac{1}{4} range$$

Example: Sample Size when Estimating μ

$$E = 1, \quad \sigma = 4$$

$$90\% \text{ confidence} \Rightarrow Z = 1.645$$

$$\begin{aligned} n &= \frac{Z_{\frac{\alpha}{2}}^2 \sigma^2}{E^2} \\ &= \frac{(1.645)^2 (4)^2}{1^2} \\ &= 43.30 \text{ or } 44 \end{aligned}$$

Example

$$E = 2, \text{ range} = 25$$

$$95\% \text{ confidence} \Rightarrow Z = 1.96$$

$$\text{estimated } \sigma: \frac{1}{4} \text{range} = \left(\frac{1}{4}\right)(25) = 6.25$$

$$\begin{aligned} n &= \frac{Z^2 \sigma^2}{E^2} \\ &= \frac{(1.96)^2 (6.25)^2}{2^2} \\ &= 37.52 \text{ or } 38 \end{aligned}$$

Determining Sample Size when Estimating P

- Z formula

$$Z = \frac{\hat{p} - P}{\sqrt{\frac{P \cdot Q}{n}}}$$

- Error of Estimation (tolerable error)

$$E = \hat{p} - P$$

- Estimated Sample Size

$$n = \frac{Z^2 PQ}{E^2}$$

Example

$$E = 0.03$$

$$98\% \text{ Confidence} \Rightarrow Z = 2.33$$

$$\text{estimated } P = 0.40$$

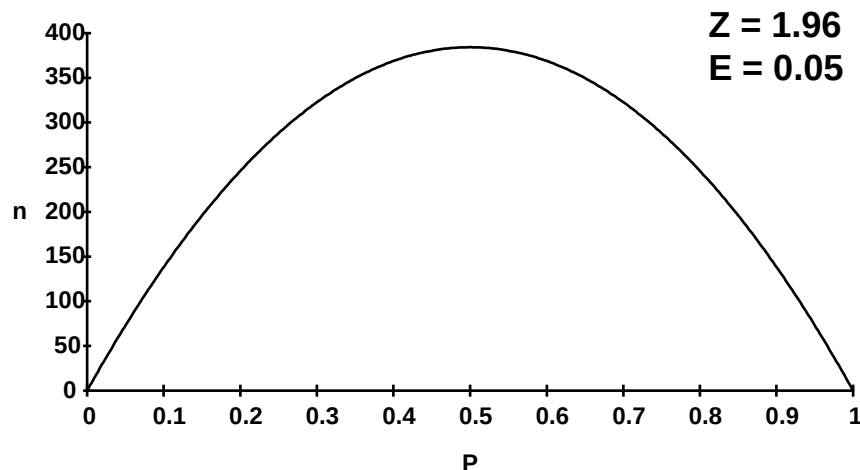
$$Q = 1 - P = 0.60$$

$$\begin{aligned} n &= \frac{Z^2 PQ}{E^2} \\ &= \frac{(2.33)^2 (0.40)(0.60)}{(.003)^2} \\ &= 1,447.7 \text{ or } 1,448 \end{aligned}$$

Determining Sample Size when Estimating P with No Prior Information

P	PQ
0.5	0.25
0.4	0.24
0.3	0.21
0.2	0.16
0.1	0.09

$$n = \frac{Z^2 \frac{1}{4}}{E^2}$$



Example

$$E = 0.05$$

$$90\% \text{ Confidence} \Rightarrow Z = 1.645$$

with no prior estimate of P , use $P = 0.50$

$$Q = 1 - P = 0.50$$

$$\begin{aligned} n &= \frac{Z^2 PQ}{E^2} \\ &= \frac{(1.645)^2 (0.50)(0.50)}{(.05)^2} \\ &= 270.6 \text{ or } 271 \end{aligned}$$

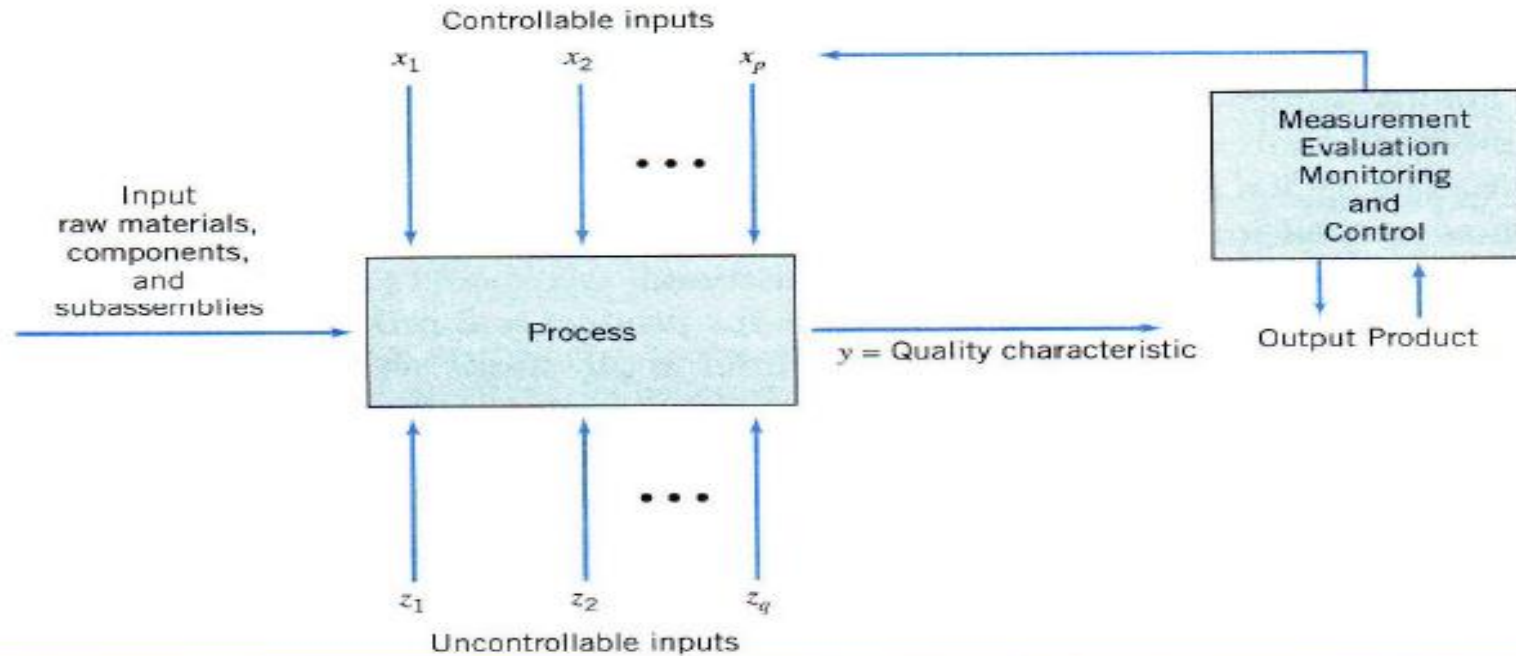
Why ANOVA?

- We could compare the means, one by one using t-tests for difference of means.
- Problem: each test contains type I error
- The total type I error is $1 - (1 - \alpha)^k$ where k is the number of means.
- For example, if there are 5 means and you use $\alpha = .05$, you must make 10 two by two comparisons.
- Thus, the type I error is $1 - (.95)^{10}$, which is .4012.
- That is, 40% of the time you will reject the null hypothesis of equal means in favor of the alternative!

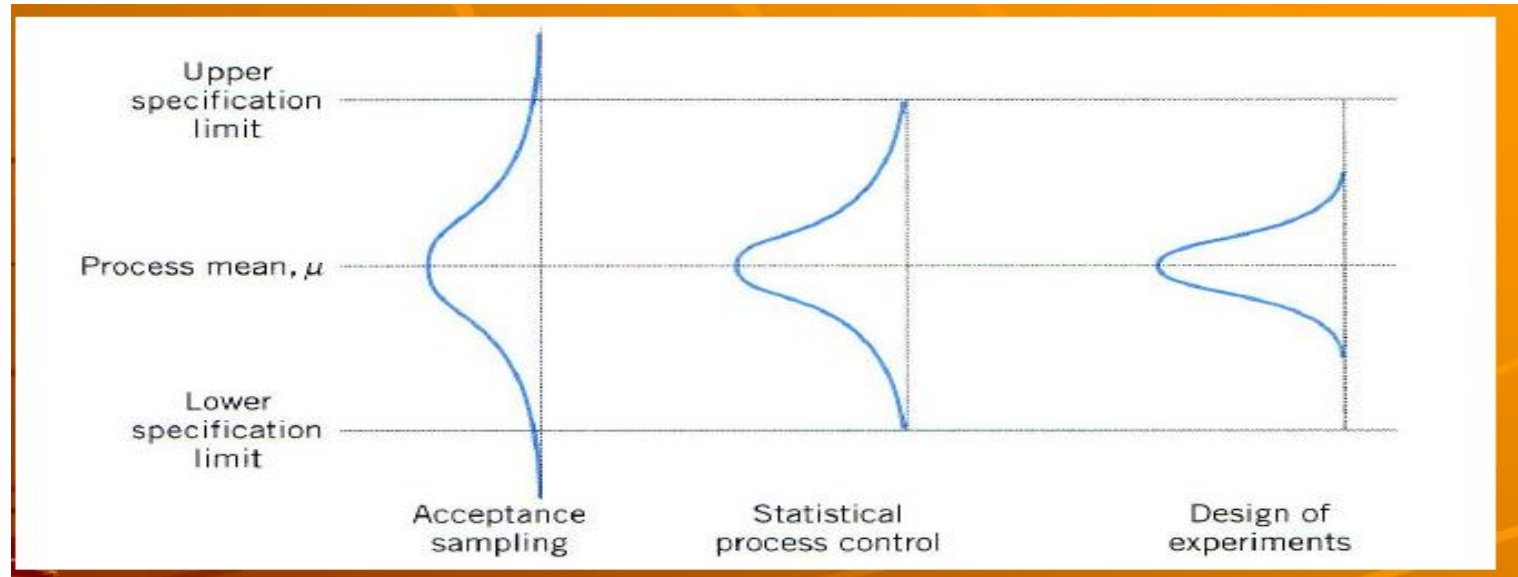
Hypothesis Testing With Categorical Data

- Chi Square tests can be viewed as a generalization of Z tests of proportions
- Analysis of Variance (ANOVA) can be viewed as a generalization of t-tests: a comparison of differences of means across more than 2 groups.
- Like Chi Square, if there are only two groups, the two analyses will produce identical results – thus a t-test or ANOVA can be used with 2 groups

Production Process inputs and outputs



Application of quality-engineering techniques and the systematic reduction of process variability



Effect of Teaching Methodology

Group 1 Black Board	Group 2 Case Presentation	Group 3 PPT
4	2	2
3	4	1
2	6	3

$$\bar{x}_1 = \frac{4+3+2}{3} = 3$$

$$\bar{x}_2 = \frac{2+4+6}{3} = 4$$

$$\bar{x}_3 = \frac{2+1+3}{3} = 2$$

$$\bar{x} = \frac{4+3+2+2+4+6+2+1+3}{9} = 3$$

$$\begin{aligned} SST &= (4-3)^2 + (3-3)^2 + (2-3)^2 + (2-3)^2 + (4-3)^2 + (6-3)^2 + (2-3)^2 + (1-3)^2 + (3-3)^2 \\ &= 1 + 0 + 1 + 1 + 1 + 9 + 1 + 4 + 0 = 18 \end{aligned}$$

$$\begin{aligned} SSB &= 3(3-3)^2 + 3(4-3)^2 + 3(2-3)^2 \\ &= 0 + 3 + 3 = 6 \end{aligned}$$

$$\begin{aligned} SSE &= (4-3)^2 + (3-3)^2 + (2-3)^2 + (2-4)^2 + (4-4)^2 + (6-4)^2 + (2-2)^2 + (1-2)^2 + (3-2)^2 \\ &= 1 + 0 + 1 + 4 + 0 + 4 + 0 + 1 + 1 = 12 \end{aligned}$$

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	6	2	3	1.5	0.296296	5.143253
Within Groups	12	6	2			
Total	18	8				

Thank You





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Effect of Teaching Methodology

Group 1 Black Board	Group 2 Case Presentation	Group 3 PPT
4	2	2
3	4	1
2	6	3

ANOVA with Python

```
In [15]: a=[4,3,2]
```

```
In [16]: b=[2,4,6]
```

```
In [17]: c=[2,1,3]
```

```
In [18]: stats.f_oneway(a,b,c)
```

```
Out[18]: F_onewayResult(statistic=1.5, pvalue=0.2962962962962962)
```

Pandas.melt command

- Pd.melt allows you to ‘unpivot’ data from a ‘wide format’ into a ‘long format’, data with each row representing a data point.

Jupyter code

```
In [22]: import pandas as pd
import numpy as np
import math
from scipy import stats
import scipy
import statsmodels.api as sm
from statsmodels.formula.api import ols
from matplotlib import pyplot as plt
```

```
In [23]: data=pd.read_excel('oneway.xlsx')
```

```
In [24]: data
```

Out[24]:

	Teachin Method1	Teachin Method2	Teachin Method3
0	4	2	2
1	3	4	1
2	2	6	3

```
In [26]: data_new=pd.melt(data.reset_index(),id_vars=['index'], value_vars=['Teachin Method1','Teachin Method2','Teachin Method3'])  
data_new.columns=['index','Treatments','value']
```

```
In [27]: data_new
```

Transforming table

```
4]: data
```

```
4]:
```

	Teachin Method1	Teachin Method2	Teachin Method3
0	4	2	2
1	3	4	1
2	2	6	3



```
In [27]: data_new
```

```
Out[27]:
```

	index	Treatments	value
0	0	Teachin Method1	4
1	1	Teachin Method1	3
2	2	Teachin Method1	2
3	0	Teachin Method2	2
4	1	Teachin Method2	4
5	2	Teachin Method2	6
6	0	Teachin Method3	2
7	1	Teachin Method3	1
8	2	Teachin Method3	3


```
In [31]: model=ols('value ~ C(Treatments)',data=data_new).fit()
```

```
In [32]: anova_table=sm.stats.anova_lm(model, typ=1)
```

```
In [33]: anova_table
```

Out[33]:

	df	sum_sq	mean_sq	F	PR(>F)
C(Treatments)	2.0	6.0	3.0	1.5	0.296296
Residual	6.0	12.0	2.0	NaN	NaN

Analysis of Variance: A Conceptual Overview

- Analysis of Variance (ANOVA) can be used to test for the equality of three or more population means
- Data obtained from observational or experimental studies can be used for the analysis
- We want to use the sample results to test the following hypotheses:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

H_a : Not all population means are equal

Analysis of Variance: A Conceptual Overview

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

H_a : Not all population means are equal

- If H_0 is rejected, we cannot conclude that all population means are equal
- Rejecting H_0 means that at least two population means have different values

Analysis of Variance: A Conceptual Overview

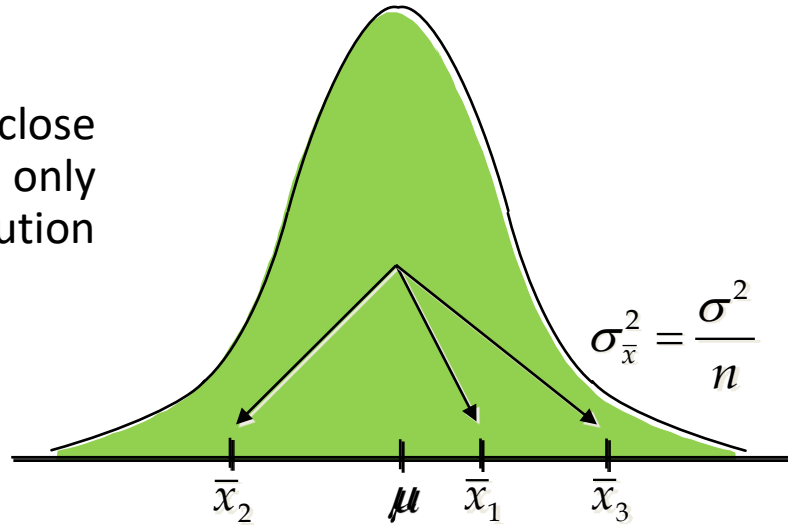
Assumptions for Analysis of Variance

- For each population, the response (dependent) variable is normally distributed
- The variance of the response variable, denoted σ^2 , is the same for all of the populations
- The observations must be independent

Analysis of Variance: A Conceptual Overview

- Sampling Distribution of $\bar{x}$ Given H_0 is True

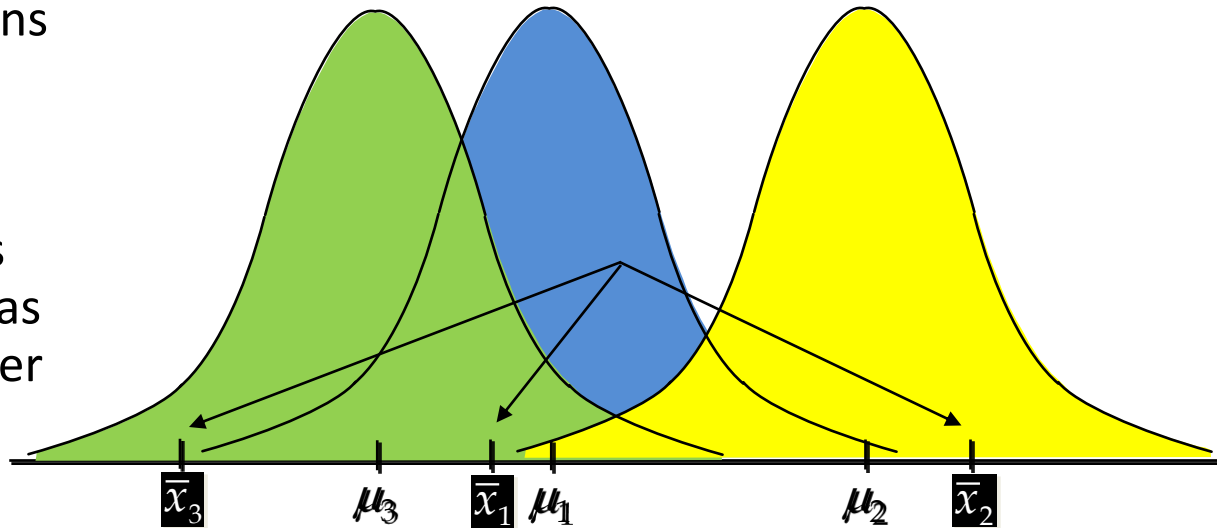
Sample means are close together because there is only one sampling distribution when H_0 is true.

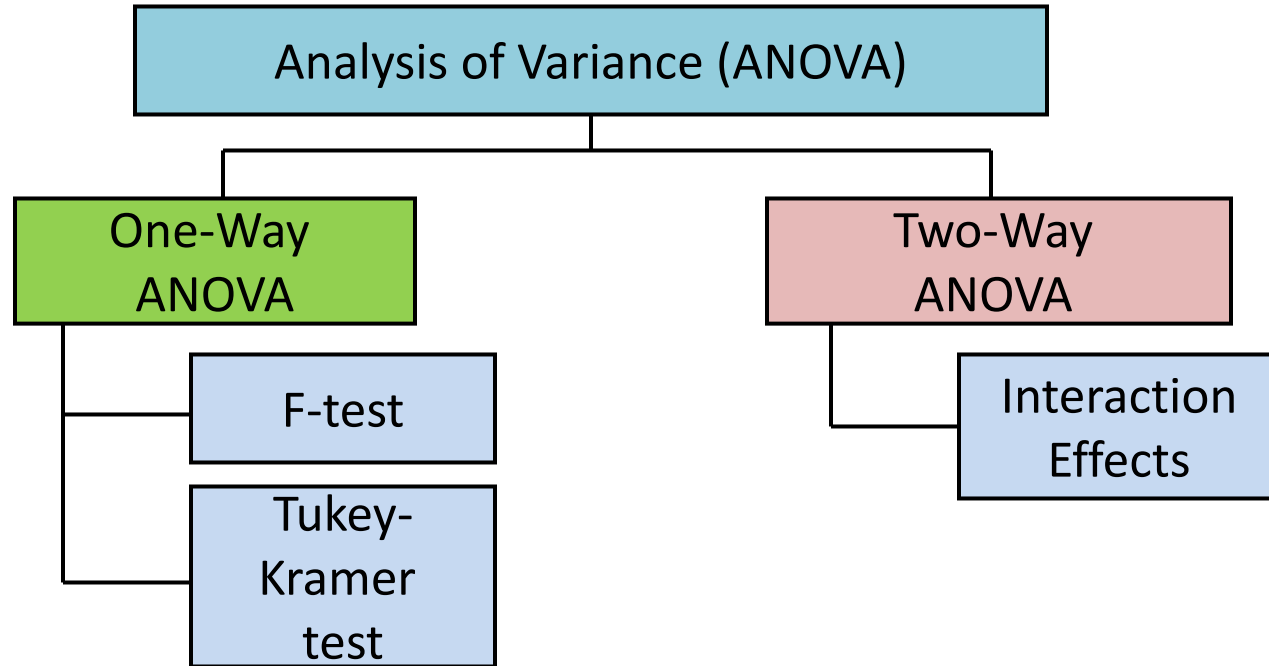


Analysis of Variance: A Conceptual Overview

- Sampling Distribution of $\bar{x}$ Given H_0 is False

Sample means come from different sampling distributions and are not as close together when H_0 is false.





General ANOVA Setting

- Investigator controls one or more factors of interest
 - Each factor contains two or more levels
 - Levels can be numerical or categorical
 - Different levels produce different groups
 - Think of the groups as populations
- Observe effects on the dependent variable
 - Are the groups the same?
- Experimental design: the plan used to collect the data



Completely Randomized Design

- Experimental units (subjects) are assigned randomly to the different levels (groups)
 - Subjects are assumed homogeneous
- Only one factor or independent variable
 - With two or more levels (groups)
- Analyzed by one-factor analysis of variance (one-way ANOVA)



Analysis of Variance and the Completely Randomized Design

- Between-Treatments Estimate of Population Variance
- Within-Treatments Estimate of Population Variance
- Comparing the Variance Estimates: The F Test
- ANOVA Table

Analysis of Variance and the Completely Randomized Design

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

H_a : Not all population means are equal

where

μ_j = mean of the j^{th} population

Analysis of Variance and the Completely Randomized Design

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

H_a : Not all population means are equal

- Assume that a simple random sample of size n_j has been selected from each of the k populations or treatments. For the resulting sample data, let x_{ij} = value of observation i for treatment j

n_j = number of observations for treatment j

$\bar{x}_j$ = sample mean for treatment j

s_j^2 = sample variance for treatment j

s_j = sample standard deviation for treatment j

Between-Treatments Estimate of Population Variance σ^2

- The estimate of σ^2 based on the variation of the sample means is called the mean square due to treatments and is denoted by MSTR

$$\text{MSTR} = \frac{\sum_{j=1}^k n_j (\bar{x}_j - \bar{\bar{x}})^2}{k-1}$$

Denominator is the degrees of freedom associated with SSTR

Numerator is called the sum of squares due to treatments (SSTR)

Between-Treatments Estimate of Population Variance σ^2

- Mean Square due to Treatments (MSTR)

$$\text{MSTR} = \frac{\sum_{j=1}^k n_j (\bar{x}_j - \bar{\bar{x}})^2}{k-1}$$

Where:

k = number of groups

n_j = sample size from group j

$\bar{x}_j$ = sample mean from group j

$\bar{\bar{x}}$ = grand mean (mean of all data values)

Within-Treatments Estimate of Population Variance σ^2

- The estimate of σ^2 based on the variation of the sample observations within each sample is called the mean square error and is denoted by MSE

$$\text{MSE} = \frac{\sum_{j=1}^k (n_j - 1)s_j^2}{n_T - k}$$

Denominator is the degrees of freedom associated with SSE

Numerator is called the sum of squares due to error (SSE)

Within-Treatments Estimate of Population Variance σ^2

- Mean Square Error (MSE)

$$\text{MSE} = \frac{\sum_{j=1}^k (n_j - 1)s_j^2}{n_T - k}$$

Where:

k = number of groups

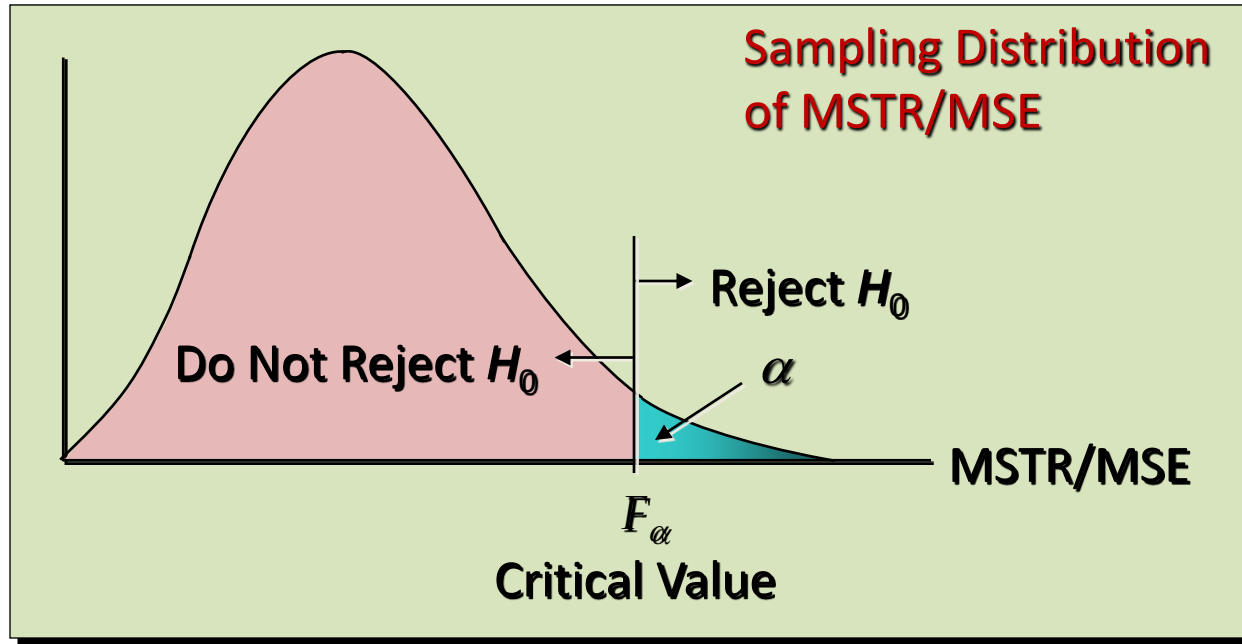
n_j = number of observations for treatment j
sample variance for treatment j

$s_j^2 =$

Comparing the Variance Estimates: The F Test

- If the null hypothesis is true and the ANOVA assumptions are valid, the sampling distribution of $MSTR/MSE$ is an F distribution with $MSTR$ d.f. equal to $k - 1$ and MSE d.f. equal to $n_T - k$.
- If the means of the k populations are not equal, the value of $MSTR/MSE$ will be inflated because $MSTR$ overestimates σ^2
- Hence, we will reject H_0 if the resulting value of $MSTR/MSE$ appears to be too large to have been selected at random from the appropriate F distribution

Comparing the Variance Estimates: The F Test



ANOVA Table for a Completely Randomized Design

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F	p -Value
Treatments	SSTR	$k - 1$	$MSTR = \frac{SSTR}{k-1}$	$\frac{MSTR}{MSE}$	
Error	SSE	$n_T - k$	$MSE = \frac{SSE}{n_T - k}$		
Total	SST	$n_T - 1$			

SST is partitioned into SSTR and SSE.

SST's degrees of freedom (d.f.) are partitioned into SSTR's d.f. and SSE's d.f.

ANOVA Table for a Completely Randomized Design

- SST divided by its degrees of freedom $n_T - 1$ is the overall sample variance that would be obtained if we treated the entire set of observations as one data set.
- With the entire data set as one sample, the formula for computing the total sum of squares, SST, is:

$$SST = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{\bar{x}})^2 = SSTR + SSE$$

ANOVA Table for a Completely Randomized Design

- ANOVA can be viewed as the process of partitioning the total sum of squares and the degrees of freedom into their corresponding sources: treatments and error
- Dividing the sum of squares by the appropriate degrees of freedom provides the variance estimates and the F value used to test the hypothesis of equal population means.

Test for the Equality of k Population Means

- Hypotheses

$$H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$

H_a : Not all population means are equal

- Test Statistic

$$F = \frac{MSTR}{MSE}$$

Test for the Equality of k Population Means

p- Value Approach

Reject H_0 if $p\text{-value} \leq \alpha$

Critical Value Approach

Reject H_0 if $F \geq F_\alpha$

Where the value of F_α is based on an F distribution with $k - 1$ numerator d.f. and $n_T - k$ denominator d.f.

Thank You





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Post Hoc Analysis(Tukey's test)

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Lecture Objectives

After completing this lecture, you should be able to:

- Use Tukey's test and LSD Test to identify specific differences between means

Designing engineering experiments

- Experimental design methods are also useful in engineering design activities, where new products are developed and existing ones are improved
- By using designed experiments, engineers can determine which subset of the process variables has the greatest influence on process performance

Designing engineering experiments

- The results of an experiment can lead to
 1. Improved process yield
 2. Reduced variability in the process and closer conformance to nominal or target requirements
 3. Reduced design and development time
 4. Reduced cost of operation

Designing engineering experiments

- Every experiment involves a sequence of activities:
 1. **Conjecture**—the original hypothesis that motivates the experiment
 2. **Experiment**—the test performed to investigate the conjecture
 3. **Analysis**—the statistical analysis of the data from the experiment
 4. **Conclusion**—what has been learned about the original conjecture from the experiment. Often the experiment will lead to a revised conjecture, and a new experiment, and so forth

The completely randomized single-factor experiment example

- A manufacturer of paper that is used for making grocery bags is interested in improving the tensile strength of the product
- Product engineer thinks that tensile strength is a function of the hardwood concentration in the pulp and that the range of hardwood concentrations of practical interest is between 5 and 20%.



The completely randomized single-factor experiment example

- A team of engineers responsible for the study decides to investigate four levels of hardwood concentration: 5%, 10%, 15%, and 20%.
- They decide to make up six test specimens at each concentration level, using a pilot plant.
- All 24 specimens are tested on a laboratory tensile tester, in random order. The data from this experiment are shown in Table

The completely randomized single-factor experiment example

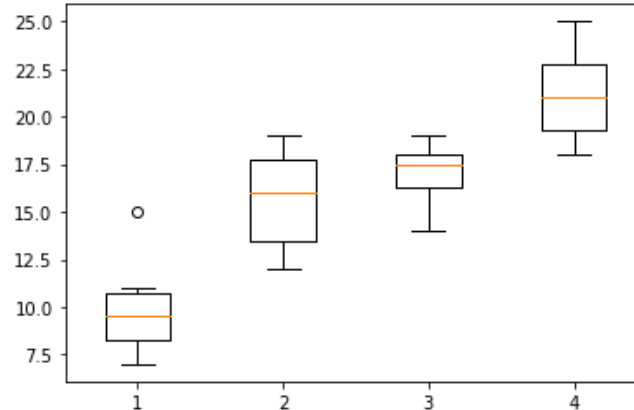
- Tensile Strength of Paper (psi)

Hardwood Concentration (%)	Observations						Total	Avg
	1	2	3	4	5	6		
5	7	8	15	11	9	10	60	10.00
10	12	17	13	18	19	15	94	15.67
15	14	18	19	17	16	18	102	17.00
20	19	25	22	23	18	20	127	21.17
							383	15.96

The completely randomized single-factor experiment example

```
In [3]: fivepercent= [7,8,15,11,9,10]
tenpercent=[12,17,13,18,19,15]
fifteenpercent=[14,18,19,17,16,18]
twentypercent=[19,25,22,23,18,20]

box_plot_data=[fivepercent,tenpercent,fifteenpercent,twentypercent]
plt.boxplot(box_plot_data)
plt.show()
```



Typical Data for Single Factor Experiment

Treatment		Observations			Totals	Averages
1	y_{11}	y_{12}	$\dots$	y_{1n}	$y_{1.}$	$\overline{y}_{1.}$
2	y_{21}	y_{23}	$\dots$	y_{2n}	$y_{2.}$	$\overline{y}_{2.}$
.	.	.	$\dots$	.	.	.
.	.	.	$\dots$	.	.	.
.	.	.	$\dots$	.	.	.
a	y_{a1}	y_{a2}	$\dots$	y_{an}	$y_{a.}$	$\overline{y}_{a.}$
					$y_{..}$	$\overline{y}_{..}$

Sum of Squares

$$\text{Total sum of squares} = SS_T = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2$$

$$\text{Treatment sum of squares} = SS_{\text{Treatments}} = n \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2$$

$$\text{Error sum of Squares} = SSE = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{j.})^2$$

ANOVA with Equal Sample Sizes

$$SST = \sum_{i=1}^a \sum_{j=1}^n y_{ij}^2 - \frac{y^2_{..}}{N}$$

$$SSTreatments = \frac{1}{n} \sum_{i=1}^a y_{i.}^2 - \frac{y^2_{..}}{N}$$

$N = an =$ No. of Treatments x no. of sample size = Total no. of Sample Size



ANOVA with unequal Sample Sizes

$$SST = \sum_{i=1}^a \sum_{j=1}^{n_i} y_{ij}^2 - \frac{y_{..}^2}{N}$$

$$SSTreatments = \sum_{i=1}^a \frac{y_{i.}^2}{n_i} - \frac{y_{..}^2}{N}$$

$N = an =$ No. of Treatments x no. of sample size = Total no. of Sample Size

Problem: Analysis of variance

- Consider the paper tensile strength experiment described.
- We can use the analysis of variance to test the hypothesis that different hardwood concentrations do not affect the mean tensile strength of the paper.
- The hypotheses are
- $H_0: \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$
- $H_1: \tau_i \neq 0$ for at least one i

Problem: Analysis of variance

- We will use $\alpha = 0.01$.
- The sums of squares for the analysis of variance are computed as follows:

$$\begin{aligned} SS_T &= \sum_{i=1}^4 \sum_{j=1}^6 y_{ij}^2 - \frac{y_{..}^2}{N} \\ &= (7)^2 + (8)^2 + \cdots + (20)^2 - \frac{(383)^2}{24} = 512.96 \end{aligned}$$

$$\begin{aligned} SS_{\text{Treatments}} &= \sum_{i=1}^4 \frac{y_{i.}^2}{n} - \frac{y_{..}^2}{N} \\ &= \frac{(60)^2 + (94)^2 + (102)^2 + (127)^2}{6} - \frac{(383)^2}{24} = 382.79 \end{aligned}$$

$$\begin{aligned} SS_E &= SS_T - SS_{\text{Treatments}} \\ &= 512.96 - 382.79 = 130.17 \end{aligned}$$

ANOVA Table

Sources of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F
Treatments	SS Treatments	$a-1$	MS Treatments	MS Treatments / MSE
Error	SSE	$a(n-1)$	MSE	
Total	SST	$an-1$		

Problem: Analysis of variance

- The ANOVA is summarized as follow

Source of Variation	Sum of Squares	Degrees of freedom	Mean Square	Fo	P-value
Hardwood concentration	382.79	3	127.6	19.6	3.59 E-6
Error	130.17	20	6.51		
Total	512.96	23			

```
In [9]: from scipy import stats
1-scipy.stats.f.cdf(19.6, 3, 20)

Out[9]: 3.599599239012541e-06
```

Problem: Analysis of variance

- Since $f_{0.01,3,20} = 4.94$, we reject H_0 and conclude that hardwood concentration in the pulp significantly affects the mean strength of the paper

```
In [32]: scipy.stats.f.ppf(1-0.01, dfn=3, dfd=20)
```

```
Out[32]: 4.938193382310539
```

Problem: Analysis of variance

```
In [23]: scipy.stats.f_oneway(fivepercent,tenpercent,fifteenpercent,twentypercent)
```

```
Out[23]: F_onewayResult(statistic=19.605206999573184, pvalue=3.5925782584743027e-06)
```

Jupyter code

```
In [1]: import pandas as pd
import numpy as np
import scipy
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

```
In [3]: df = pd.read_excel('concentration.xlsx')
df
```

Out[3]:

	concentration5	concentration10	concentration15	concentration20
0	7	12	14	19
1	8	17	18	25
2	15	13	19	22
3	11	18	17	23
4	9	19	16	18
5	10	15	18	20

Jupyter code

```
In [5]: data_r1 = pd.melt(df.reset_index(), id_vars=['index'], value_vars=['concentration5', 'concentration10', 'concentration15', 'concentration20'])  
data_r1.columns = ['index', 'treatments', 'value']
```

```
In [6]: model = ols('value ~ C(treatments)', data=data_r1).fit()
```

```
In [7]: model.summary()
```

Out[7]:

Jupyter code

Out[7]: OLS Regression Results

Dep. Variable:	value	R-squared:	0.746
Model:	OLS	Adj. R-squared:	0.708
Method:	Least Squares	F-statistic:	19.61
Date:	Tue, 27 Aug 2019	Prob (F-statistic):	3.59e-06
Time:	15:03:38	Log-Likelihood:	-54.344
No. Observations:	24	AIC:	116.7
Df Residuals:	20	BIC:	121.4
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	15.6667	1.041	15.042	0.000	13.494	17.839
C(treatments)[T.concentration15]	1.3333	1.473	0.905	0.376	-1.739	4.406
C(treatments)[T.concentration20]	5.5000	1.473	3.734	0.001	2.428	8.572
C(treatments)[T.concentration5]	-5.6667	1.473	-3.847	0.001	-8.739	-2.594

Omnibus:	0.929	Durbin-Watson:	2.181
Prob(Omnibus):	0.628	Jarque-Bera (JB):	0.861
Skew:	0.248	Prob(JB):	0.650
Kurtosis:	2.215	Cond. No.	4.79

Jupyter code

```
In [8]: aov_table = sm.stats.anova_lm(model, typ=1)  
aov_table
```

Out[8]:

	df	sum_sq	mean_sq	F	PR(>F)
C(treatments)	3.0	382.791667	127.597222	19.605207	0.000004
Residual	20.0	130.166667	6.508333	NaN	NaN

Multiple Comparisons Following the ANOVA

- When the null hypothesis is rejected in the ANOVA, we know that some of the treatment or factor level means are different
- ANOVA doesn't identify which means are different
- Methods for investigating this issue are called multiple comparisons methods

Fisher's least significant difference (LSD) method

- The Fisher LSD method compares all pairs of means with the null hypotheses $H_0: \mu_i = \mu_j$ (for all $i \neq j$) using the t-statistic

$$t_0 = \frac{\overline{y_{i^*}} - \overline{y_{j^*}}}{\sqrt{\frac{2MS_E}{n}}}$$

Fisher's least significant difference (LSD) method

- Assuming a two-sided alternative hypothesis, the pair of means i and j would be declared significantly different if

$$\left| \overline{y}_{i^*} - \overline{y}_{j^*} \right| > LSD$$

where LSD, the least significant difference, is

$$LSD = t_{\alpha/2, a(n-1)} \sqrt{\frac{2MS_E}{n}}$$

Fisher's least significant difference (LSD) method

- If the sample sizes are different in each treatment, the LSD is defined as

$$LSD = t_{\alpha/2, N-a} \sqrt{MS_E \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

Problem : LSD method

- We will apply the Fisher LSD method to the hardwood concentration experiment. There are $a = 4$ means, $n = 6$, $MSE = 6.51$, and $t_{0.025,20} = 2.086$. The treatment means are

$$\bar{y}_{1\cdot} = 10.00 \text{ psi}$$

$$\bar{y}_{2\cdot} = 15.67 \text{ psi}$$

$$\bar{y}_{3\cdot} = 17.00 \text{ psi}$$

$$\bar{y}_{4\cdot} = 21.17 \text{ psi}$$

Problem : LSD method

- The value of LSD is:

$$LSD = t_{0.025,20} \sqrt{\frac{2MS_E}{n}} = 2.086 \sqrt{\frac{2(6.51)}{6}} = 3.07$$

- Therefore, any pair of treatment averages that differs by more than 3.07 implies that the corresponding pair of treatment means are different.

Jupyter code

```
In [8]: aov_table = sm.stats.anova_lm(model, typ=1)
aov_table
```

Out[8]:

	df	sum_sq	mean_sq	F	PR(>F)
C(treatments)	3.0	382.791667	127.597222	19.605207	0.000004
Residual	20.0	130.166667	6.508333	NaN	NaN

LSD

```
In [39]: import math
t = -1*scipy.stats.t.ppf(0.025,20)
n= 6
MSE = 6.508333
lsd = t*math.sqrt(2*MSE/n)
lsd
```

Out[39]: 3.072422588325206

Problem : LSD method

- The comparisons among the observed treatment averages are as follows:

$$4 \text{ vs. } 1 = 21.17 - 10.00 = 11.17 > 3.07$$

$$4 \text{ vs. } 2 = 21.17 - 15.67 = 5.50 > 3.07$$

$$4 \text{ vs. } 3 = 21.17 - 17.00 = 4.17 > 3.07$$

$$3 \text{ vs. } 1 = 17.00 - 10.00 = 7.00 > 3.07$$

$$3 \text{ vs. } 2 = 17.00 - 15.67 = 1.33 < 3.07$$

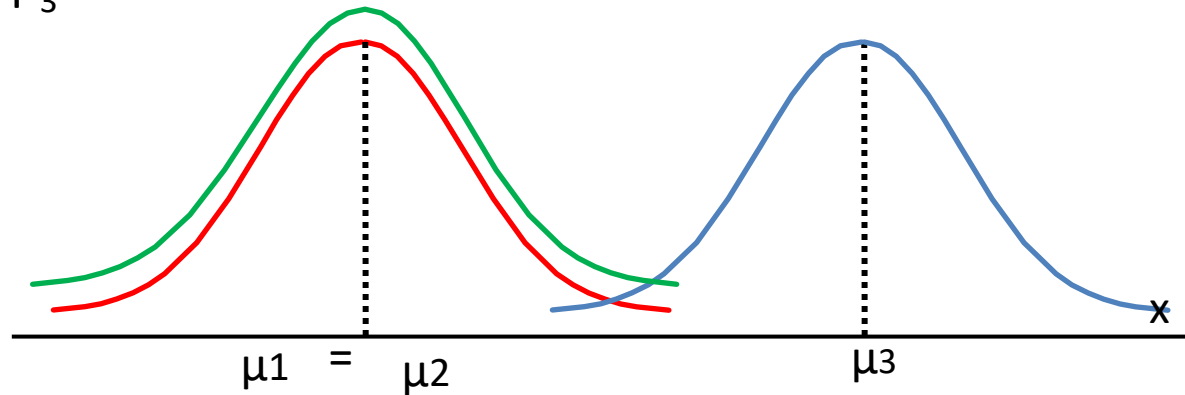
$$2 \text{ vs. } 1 = 15.67 - 10.00 = 5.67 > 3.07$$

The Tukey-Kramer Test for Post Hoc analysis

- Tells which population means are significantly different
- Done after rejection of equal means in ANOVA
- Allows pair-wise comparisons
- Compare absolute mean differences with critical range

The Tukey-Kramer Test for Post Hoc analysis

- Determine if there is any significant difference between the means
- is $\mu_1 = \mu_2 \neq \mu_3$



Tukey-Kramer Critical Range

$$\text{Critical Range} = Q_U \sqrt{\frac{\text{MSW}}{2} \left(\frac{1}{n_j} + \frac{1}{n_{j'}} \right)}$$

where:

Q_U = Value from Studentized Range

Distribution with c and $n - c$ degrees of freedom for
the desired level of α

MSW = Mean Square Within

n_j and $n_{j'}$ = Sample sizes from groups j and j'

Problem: Tukey- Kramer test

- Tensile Strength of Paper (psi)

Hardwood Concentratio n (%)	Observations						Total	Avg
	1	2	3	4	5	6		
5	7	8	15	11	9	10	60	10.00
10	12	17	13	18	19	15	94	15.67
15	14	18	19	17	16	18	102	17.00
20	19	25	22	23	18	20	127	21.17
							383	15.96

The Tukey-Kramer Procedure

1. Compute absolute mean differences:

$$|\bar{x}_1 - \bar{x}_2| = |10.00 - 15.67| = 5.67$$

$$|\bar{x}_1 - \bar{x}_3| = |10.00 - 17.00| = 7$$

$$|\bar{x}_2 - \bar{x}_3| = |15.67 - 17.00| = 1.33$$

$$|\bar{x}_1 - \bar{x}_4| = |10.00 - 21.17| = 11.17$$

$$|\bar{x}_2 - \bar{x}_4| = |15.67 - 21.17| = 5.5$$

$$|\bar{x}_3 - \bar{x}_4| = |17.00 - 21.17| = 4.17$$

The Tukey-Kramer Procedure

2. Find the Q_U value from the table with $c = 4$ and $(n - c) = (24 - 4) = 20$ degrees of freedom for the desired level of α ($\alpha = .05$ used here):

$$Q_U = 3.96$$

Error Term	2	3	4	5	6	7	8	9	10
5	3.64 5.70	4.60 6.98	5.22 7.80	5.67 8.42	6.03 8.91	6.33 9.32	6.58 9.67	6.80 9.97	6.99 10.24
6	3.46 5.24	4.34 6.33	4.90 7.03	5.30 7.56	5.63 7.97	5.90 8.32	6.12 8.61	6.32 8.87	6.49 9.10
7	3.34 4.95	4.16 5.92	4.68 6.54	5.06 7.01	5.36 7.37	5.61 7.68	5.82 7.94	6.00 8.17	6.16 8.37
8	3.26 4.75	4.04 5.64	4.53 6.20	4.89 6.62	5.17 6.96	5.40 7.24	5.60 7.47	5.77 7.68	5.92 7.86
9	3.20 4.60	3.95 5.43	4.41 5.96	4.76 6.35	5.02 6.66	5.24 6.91	5.43 7.13	5.59 7.33	5.74 7.49
10	3.15 4.48	3.88 5.27	4.33 5.77	4.65 6.14	4.91 6.43	5.12 6.67	5.30 6.87	5.46 7.05	5.60 7.21
11	3.11 4.39	3.82 5.15	4.26 5.62	4.57 5.97	4.82 6.25	5.03 6.48	5.20 6.67	5.35 6.84	5.49 6.99
12	3.08 4.32	3.77 5.05	4.20 5.50	4.51 5.84	4.75 6.10	4.95 6.32	5.12 6.51	5.27 6.67	5.39 6.81
13	3.06 4.26	3.73 4.96	4.15 5.40	4.45 5.73	4.69 5.98	4.88 6.19	5.05 6.37	5.19 6.53	5.32 6.67
14	3.03 4.21	3.70 4.89	4.11 5.32	4.41 5.63	4.64 5.88	4.83 6.08	4.99 6.26	5.13 6.41	5.25 6.54
15	3.01 4.17	3.67 4.84	4.08 5.25	4.37 5.56	4.59 5.80	4.78 5.99	4.94 6.16	5.08 6.31	5.20 6.44
16	3.00 4.13	3.65 4.79	4.05 5.19	4.33 5.49	4.56 5.72	4.74 5.92	4.90 6.08	5.03 6.22	5.15 6.35
17	2.98 4.10	3.63 4.74	4.02 5.14	4.30 5.43	4.52 5.66	4.70 5.85	4.86 6.01	4.99 6.15	5.11 6.27
18	2.97 4.07	3.61 4.70	4.00 5.09	4.28 5.38	4.49 5.60	4.67 5.79	4.82 5.94	4.96 6.08	5.07 6.20
19	2.96 4.05	3.59 4.67	3.98 5.07	4.25 5.33	4.47 5.55	4.65 5.73	4.79 5.89	4.92 6.02	5.04 6.14
20	2.95 4.02	3.58 4.64	3.96 5.05	4.23 5.29	4.45 5.51	4.62 5.69	4.77 5.84	4.90 5.97	5.01 6.09

Q table: The critical values for q corresponding to $\alpha = .05$ (top) and $\alpha = .01$ (bottom)

The Tukey-Kramer Procedure

Source of Variation	Sum of Squares	Degrees of freedom	Mean Square	F ₀	P-value
Hardwood concentration	382.79	3	127.6	19.6	3.59 E-6
Error	130.17	20	6.51		
Total	512.96	23			

The Tukey-Kramer Procedure

3. Compute Critical Range:

$$\text{Critical Range} = Q_U \sqrt{\frac{MSW}{2} \left(\frac{1}{n_j} + \frac{1}{n_{j'}} \right)} = 3.96 \sqrt{\frac{6.51}{2} \left(\frac{1}{6} + \frac{1}{6} \right)} = 4.124$$

4. Compare:

$$|\bar{x}_1 - \bar{x}_2| = |10.00 - 15.67| = 5.67$$

$$|\bar{x}_1 - \bar{x}_3| = |10.00 - 17.00| = 7$$

$$|\bar{x}_2 - \bar{x}_3| = |15.67 - 17.00| = 1.33$$

$$|\bar{x}_1 - \bar{x}_4| = |10.00 - 21.17| = 11.17$$

$$|\bar{x}_2 - \bar{x}_4| = |15.67 - 21.17| = 5.5$$

$$|\bar{x}_3 - \bar{x}_4| = |17.00 - 21.17| = 4.17$$

The Tukey-Kramer Procedure

5. Other than $|\bar{x}_2 - \bar{x}_3|$, all of the absolute **mean differences are greater than critical range**. Therefore there is significant difference between each pair of means, except 10% concentration and 15% concentration at the 5% level of significance.

Jupyter code

```
In [53]: from statsmodels.stats.multicomp import pairwise_tukeyhsd
from statsmodels.stats.multicomp import MultiComparison
mc = MultiComparison(data_r1['value'], data_r1['treatments'])
mcresult = mc.tukeyhsd(0.05)
mcresult.summary()
```

Out[53]: Multiple Comparison of Means - Tukey HSD,FWER=0.05

group1	group2	meandiff	lower	upper	reject
concentration10	concentration15	1.3333	-2.7894	5.4561	False
concentration10	concentration20	5.5	1.3773	9.6227	True
concentration10	concentration5	-5.6667	-9.7894	-1.5439	True
concentration15	concentration20	4.1667	0.0439	8.2894	True
concentration15	concentration5	-7.0	-11.1227	-2.8773	True
concentration20	concentration5	-11.1667	-15.2894	-7.0439	True

Problem 2

- Following table shows observed tensile strength (lb/in square) of different clothes having different weight percentage of cotton.
- Check whether having different weight percentage of cotton, plays any role in tensile strength (lb/in square) of clothes.



Problem 2

Weight Percentage of cotton	Observed tensile strength (lb/in square)					Total	Average
	1	2	3	4	5		
15	7	7	15	11	9	49	9.8
20	12	17	12	18	18	77	15.4
25	14	18	18	19	19	88	17.6
30	19	25	22	19	23	108	21.6
35	7	10	11	15	11	54	10.8
						Grand total=376	Grand mean=15.004

- $$SSA = 5 (9.8 - 15.04)^2 + 5 (15.4 - 15.04)^2 + 5 (17.6 - 15.04)^2 + 5 (21.6 - 15.04)^2 + 5 (10.8 - 15.04)^2 = 475.76$$

$$SST = 636.96$$

$$SSE = 636.96 - 475.76 = 161.20$$

Sources of variation	Sum of squares	Degrees of freedom	Mean square	F-value
Cotton weight percentage	475.76	4	118.94	14.76
Error	161.20	20	8.06	
Total	639.96	24		

Problem 2

- When $\alpha = .05$, $F_{(0.05, 4, 20)} = 2.87$
- Reject H_0

```
In [17]: scipy.stats.f.ppf(1-0.05, dfn=4, dfd=20)
```

```
Out[17]: 2.8660814020156584
```

Error Term	2	3	4	5	6	7	8	9	10
5	3.64 5.70	4.60 6.98	5.22 7.80	5.67 8.42	6.03 8.91	6.33 9.32	6.58 9.67	6.80 9.97	6.99 10.24
6	3.46 5.24	4.34 6.33	4.90 7.03	5.30 7.56	5.63 7.97	5.90 8.32	6.12 8.61	6.32 8.87	6.49 9.10
7	3.34 4.95	4.16 5.92	4.68 6.54	5.06 7.01	5.36 7.37	5.61 7.68	5.82 7.94	6.00 8.17	6.16 8.37
8	3.26 4.75	4.04 5.64	4.53 6.20	4.89 6.62	5.17 6.96	5.40 7.24	5.60 7.47	5.77 7.68	5.92 7.86
9	3.20 4.60	3.95 5.43	4.41 5.96	4.76 6.35	5.02 6.66	5.24 6.91	5.43 7.13	5.59 7.33	5.74 7.49
10	3.15 4.48	3.88 5.27	4.33 5.77	4.65 6.14	4.91 6.43	5.12 6.67	5.30 6.87	5.46 7.05	5.60 7.21
11	3.11 4.39	3.82 5.15	4.26 5.62	4.57 5.97	4.82 6.25	5.03 6.48	5.20 6.67	5.35 6.84	5.49 6.99
12	3.08 4.32	3.77 5.05	4.20 5.50	4.51 5.84	4.75 6.10	4.95 6.32	5.12 6.51	5.27 6.67	5.39 6.81
13	3.06 4.26	3.73 4.96	4.15 5.40	4.45 5.73	4.69 5.98	4.88 6.19	5.05 6.37	5.19 6.53	5.32 6.67
14	3.03 4.21	3.70 4.89	4.11 5.32	4.41 5.63	4.64 5.88	4.83 6.08	4.99 6.26	5.13 6.41	5.25 6.54
15	3.01 4.17	3.67 4.84	4.08 5.25	4.37 5.56	4.59 5.80	4.78 5.99	4.94 6.16	5.08 6.31	5.20 6.44
16	3.00 4.13	3.65 4.79	4.05 5.19	4.33 5.49	4.56 5.72	4.74 5.92	4.90 6.08	5.03 6.22	5.15 6.35
17	2.98 4.10	3.63 4.74	4.02 5.14	4.30 5.43	4.52 5.66	4.70 5.85	4.86 6.01	4.99 6.15	5.11 6.27
18	2.97 4.07	3.61 4.70	4.00 5.09	4.28 5.38	4.49 5.60	4.67 5.79	4.82 5.94	4.96 6.08	5.07 6.20
19	2.96 4.05	3.59 4.67	3.98 5.05	4.25 5.35	4.47 5.55	4.65 5.73	4.79 5.89	4.92 6.02	5.04 6.14
20	2.95 4.02	3.58 4.64	3.96 5.02	4.23 5.31	4.45 5.51	4.62 5.69	4.77 5.84	4.90 5.97	5.01 6.09

Q table: The critical values for q corresponding to $\alpha = .05$ (top) and $\alpha = .01$ (bottom)

Problem 2

$$T_{\alpha} = q_{\alpha}(c, n - c) \sqrt{\frac{MS_E}{n}}$$

$$\alpha = 0.05$$

$$q_{0.05}(5, 20) = 4.23$$

$$T_{0.05} = 4.23 \sqrt{\frac{8.06}{5}} = 5.37$$

Problem 2

Any pair of treatment averages that differ in absolute value by **more than 5.37** would imply that the corresponding pair of population means are significantly different.

Problem 2

$$\bar{y}_1 - \bar{y}_2 = |9.8 - 15.4| = 5.6^*$$

$$\bar{y}_1 - \bar{y}_3 = |9.8 - 17.6| = 7.8^*$$

$$\bar{y}_1 - \bar{y}_4 = |9.8 - 21.6| = 11.8^*$$

$$\bar{y}_1 - \bar{y}_5 = |9.8 - 10.8| = 1$$

Starred values indicate pairs of means that are significantly different.

$$\bar{y}_2 - \bar{y}_3 = |15.4 - 17.6| = 2.2$$

$$\bar{y}_2 - \bar{y}_4 = |15.4 - 21.6| = 6.2^*$$

$$\bar{y}_2 - \bar{y}_5 = |15.4 - 10.8| = 4.6$$

$$\bar{y}_3 - \bar{y}_4 = |17.6 - 21.6| = 4$$

$$\bar{y}_3 - \bar{y}_5 = |17.6 - 10.8| = 6.8^*$$

$$\bar{y}_4 - \bar{y}_5 = |21.6 - 10.8| = 10.8^*$$

Jupyter code

```
In [2]: df3 = pd.read_excel('C:/Users/Somi/Documents/cotton weight.xlsx')
```

```
In [12]: data1 = pd.melt(df3.reset_index(), id_vars=['index'], value_vars=['cotwt.15', 'cotwt.20', 'cotwt.25', 'cotwt.30', 'cotwt.35'])  
data1.columns = ['id', 'treatments', 'value']
```

Jupyter Code

```
In [16]: mc = MultiComparison(data1['value'], data1['treatments'])  
mcresults = mc.tukeyhsd(0.05)  
mcresults.summary()
```

Out[16]: Multiple Comparison of Means - Tukey HSD,FWER=0.05

group1	group2	meandiff	lower	upper	reject
cotwt.15	cotwt.20	5.6	0.2266	10.9734	True
cotwt.15	cotwt.25	7.8	2.4266	13.1734	True
cotwt.15	cotwt.30	11.8	6.4266	17.1734	True
cotwt.15	cotwt.35	1.0	-4.3734	6.3734	False
cotwt.20	cotwt.25	2.2	-3.1734	7.5734	False
cotwt.20	cotwt.30	6.2	0.8266	11.5734	True
cotwt.20	cotwt.35	-4.6	-9.9734	0.7734	False
cotwt.25	cotwt.30	4.0	-1.3734	9.3734	False
cotwt.25	cotwt.35	-6.8	-12.1734	-1.4266	True
cotwt.30	cotwt.35	-10.8	-16.1734	-5.4266	True

Thank you

